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Ain't Behavin': Forecast Errors and Measurement Errors in Early GNP Estimates

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Revisions of the early GNP estimates may contain elements of measurement errors as well as forecast errors. These types of error behave differently but need to satisfy a common set of criteria for well-behavedness. This article tests these criteria for U.S. GNP revisions. The tests are similar to tests of rationality and are based on the generalized method of moments estimator. The flash, 15-day, and 45-day estimates are found to be ill behaved, but the 75-day estimate satisfies the criteria for well-behavedness.

KEY WORDS: Data revisions; Efficient forecasts; Generalized method of moments.

1. INTRODUCTION

The United States probably has the world's timeliest system when it comes to the release of quarterly gross national product (GNP) estimates. Through 1985, a government agency, the Bureau of Economic Analysis (BEA), announced the so-called "flash" estimate, also known as the "minus 15-day" estimate, even two weeks before the quarter was over. Revised estimates, known as the 15-day, the 45-day, and the 75-day estimates, followed rapidly at one-month intervals. Although releases of the flash estimate now have been discontinued, the remaining estimates are reported regularly in the daily press as well as in government publications.

These early estimates are important. The financial markets react immediately to their releases. Business decisions are influenced by them, and government policy decisions are guided by them. Given this importance, an investigation of how they behave and what they tell should be of considerable interest.

This article presents the results of such an investigation. In particular, it investigates the statistical properties of the preliminary estimates of real GNP growth and their subsequent revisions. Tests are carried out to determine whether the estimates satisfy the criteria of efficient forecasts or observations with random measurement errors.

Unfortunately, the results are mostly negative. Tests carried out for the available data from 1968 through 1984 indicate that, except for the 75-day estimate, all of the preliminary estimates behave neither as efficient forecasts nor as observations with random errors. They tend to be downward biased and to excessively reflect simple extrapolations of recent growth rates. The 75-day estimate is an exception. It is well behaved; however, the validity of this conclusion is limited by the extent of the data—the 75-day estimate was not issued

regularly until 1974. The 1985 benchmark revision was not available at the time the research was done. Tests of these revisions, perhaps using a new 1972-based series currently in preparation at the BEA, seem an interesting area of further research.

The BEA carries out its own studies of GNP revisions. The data used for this article were prepared for one such study—Parker (1984). Other BEA studies have included Jaszi (1965) and Young (1974). Outside studies of data revisions have been partly descriptive (Cole 1969a; Zellner 1958), and have focused partly on the use of preliminary data for estimation and forecasting (Cole 1969b; Harvey, McKenzie, Blake, and Desai 1983; Howrey 1978, 1984).

Although I am not familiar with any studies dealing directly with the effects of GNP announcements, the parallel literature for money-supply announcement is obviously relevant (Cornell 1983; Urich 1982; Urich and Wachtel 1984). Another parallel is the so-called news models for exchange-rate movements that use announcements of balance-of-payments data (Frenkel 1981). The significance of preliminary announcements also is related to the analysis of unperceived, rather than unanticipated, money-supply changes (Barro and Hercowitz 1980; Boschen and Grossman 1982; King 1981).

Two recent articles have sought answers to questions similar to the ones asked in this article. Preliminary money-supply figures and their subsequent revisions were the subject of a study by Mankiw, Runkle, and Shapiro (1984). They concluded that money-supply revisions behave like measurement errors rather than efficient forecast errors. In a follow-up study, Mankiw and Shapiro (1986) applied similar methods to GNP data and concluded that they behave like efficient forecasts. This article draws on the methodology of Mankiw et al. (1984) but generalized it to account for cases other

than the polar cases of forecasting and measurement with error.

My findings are much less encouraging than those of Mankiw and Shapiro (1986). Some of the differences may be ascribed to their use of a shorter sample and their failure to use some of the variables used in my study. Much more important, however, is their failure to correct for biases in the estimated standard errors. Since the biases mostly go in the upward direction, my tests have much more power.

The testing model has predetermined rather than exogenous variables on the right side, so the correct treatment of serial correlation is somewhat tricky. The analysis uses the generalized method of moments (GMM) approach suggested by Hansen (1982).

The results of the article easily can be interpreted as criticism of the BEA. If any such criticism is intended, it is meant to be of the constructive kind. The main purpose of the article, however, is not to criticize but to help the users outside the BEA understand what they are dealing with when they use preliminary GNP data. The article seems to throw considerable light on this issue.

The next section summarizes the publication schedule for preliminary and revised quarterly GNP estimates in the United States. Readers familiar with this schedule may skip directly to Section 3, and those interested in a more detailed account are referred to Parker (1984), from which this section draws heavily. Section 3 presents the model used in the article, Section 4 discusses the empirical implementation of the model and the estimation problems involved, and Section 5 introduces the data. Section 6 presents the results, and Section 7 offers some comments and conclusions.

2. THE PUBLICATION SCHEDULE FOR QUARTERLY GNP DATA

Through the fourth quarter of 1985, the BEA released the so-called flash (or minus 15-day) estimate two weeks before the end of the quarter. This estimate was released only for total GNP (nominal and real), although estimates of the major components were circulated internally within the federal government. The observations for this estimate were necessarily incomplete for all components; therefore, the estimate contained a substantial amount of projection. Its observation basis consisted of preliminary, small-sample surveys for one or two months of the quarter.

Observations for the flash estimate are included in the analysis in this article. Currently, however, the first estimate to be released is the 15-day estimate, prepared a month after the time the flash estimate used to be released. As more source data become available, routine revisions lead to releases of new estimates with one-month intervals, referred to as the 45-day and the 75-day estimates. For the 15-day estimate, total GNP is released with all of the major components except for

corporate profits and related aggregates, which are released for the first time together with the 45-day estimate. By the time of the 75-day estimate, three monthly observations or one quarterly observation are available for all GNP components, except for a small proportion of business inventories and consumer services for which quarterly observations actually never become available. (The final quarterly data for these components are, in fact, interpolations of the annual data.)

After these four initial estimates, comprehensive revisions for the past three calendar years usually are published in July of each year. These revisions are based on more comprehensive annual surveys for all sectors of the economy. Along with each July revision, the BEA issues a revised estimate (the 105-day estimate) for the first quarter of the year the revision is made.

The most comprehensive revision is the benchmark revision carried out every five years. What distinguishes this revision from the others is its use of the detailed input-output tables, which are constructed from economic census data every five years. Part of the revision consists of interpolation between the census years or extrapolation from the last available census year. The input-output data become available after a substantial lag. Thus the 1985 benchmark revision made use of the 1977 input-output tables for the first time. All nominal figures from 1973 on were revised. The constant-dollar figures were revised further back as a result of the change in evaluation year from 1972 to 1982.

Changes in classifications, definitions, and other methods occur from time to time, leading to additional revisions of the historical data. For comparability, the data used in this article are normalized with respect to these changes.

3. A TESTING MODEL

Let y_t denote the growth rate of real GNP in quarter t . At the time the preliminary estimate of y_t is computed, preliminary observations for some of its components (or for parts of the quarter) are available. Let q_t denote the average growth rate of these components and z_t the corresponding rate for the remaining ones. Then,

$$y_t = kq_t + (1 - k)z_t, \quad 0 \leq k \leq 1. \quad (1)$$

The weight k is a function of the government's decisions about which surveys are to be conducted and when. I treat it as predetermined. The data presented by Parker (1984, table 1) indicate values of k of .45, .64, .74, and .83 for the flash, 15-day, 45-day, and 75-day estimates, respectively.

Let y_t^* , q_t^* , and z_t^* denote the final values of the three variables, respectively. At the time of the preliminary estimate, a preliminary observation of q_t , denoted by q_t^o , is available, but z_t is unobserved at this stage. Additional information available at the same time is denoted by X_t . I hypothesize that the measurement error v_t contained in q_t^o has a zero mean and constant vari-

ance, and that it is purely random in the sense of being uncorrelated with q_i^* and X_i :

$$q_i^o = q_i^* + v_i, \quad E(v_i) = 0, \quad v_i \perp q_i^*, \quad v_i \perp X_i. \quad (2)$$

The measurement error may arise, for example, from the use of small-sample preliminary surveys.

The preliminary observation q_i^o is not, in general, the best forecast for q_i^* . Suppose X_i can be used to forecast q_i^* unconditionally—that is, in the absence of q_i^o . Then the best conditional forecast (given q_i^o) is a function of both X_i and q_i^o . Denote it by q_i^f and note that $q_i^* - q_i^f \perp X_i$. For the unobserved components, the BEA has to rely on projections. Again, the best conditional forecast is a function of q_i^o and X_i . Denote it by z_i^f and note that $z_i^* - z_i^f \perp X_i$.

The data-collecting agency, in this case the BEA, faces the task of combining the preliminary estimate for q with the various projections to come up with a preliminary estimate y^p . In practice, the BEA may or may not estimate y^p component by component. In principle, however, y^p may be decomposed as in Equation (1). The problem of deriving y_i^p thus may be cast in terms of choosing values for q_i^p and z_i^p .

For z_i^p , since no observations are available, the BEA must use projections; that is, $z_i^p = z_i^f$. For q_i^p , the BEA has the choice between using q_i^o as the preliminary estimate as it is or using the conditional forecast q_i^f . A priori, neither alternative is superior. In fact, any weighted average of q_i^o and q_i^f could be an acceptable choice. Since the outcome of this choice is unknown to the outside observer, it seems reasonable to specify that

$$q_i^p = \lambda q_i^o + (1 - \lambda)q_i^f, \quad (3)$$

where $0 \leq \lambda \leq 1$ is an unknown parameter.

This model implicitly assumes that the forecasting activities of the BEA seek to minimize a symmetric (e.g., quadratic) loss function. The interpretation of the empirical results toward the end of the article suggests that the objectives of the BEA may be more complex. Since forecasting usually is thought of in the context of a symmetric loss function, however, it seems useful to develop and test the model under this assumption and to derive alternative suggestions from the results.

To the econometrician, only y_i^* , y_i^p , and X_i are observable. For testing the model, consider the revision $r_i = y_i^* - y_i^p$. Since Equation (1) must hold for the final values as well as the preliminary estimates, (3) implies that

$$r_i = k\lambda(q_i^* - q_i^o) + k(1 - \lambda)(q_i^* - q_i^f) + (1 - k)(z_i^* - z_i^f).$$

From (2) and the preceding arguments about efficient forecasts, it is easy to see that the revision r_i is a linear combination of random variables that all are hypothesized to be uncorrelated with the information X_i available at the time the preliminary estimate was formed.

Thus the equation

$$r_i = X_i\beta + u_i \quad (4)$$

should obey the orthogonality constraint $\beta = 0$.

It is important to note that the test of this constraint is independent of the values of λ and k . If it can be assumed that all preliminary estimates are intended to be forecasts so that $\lambda = 0$, the orthogonality test is a test of the efficiency of these forecasts. If preliminary observations are available for all components ($k = 1$) and these observations are used as the preliminary estimates so that $\lambda = 1$, the test is for the randomness of the measurement error v_i . In the general case, the orthogonality test jointly considers the efficiency of the forecast and the randomness of the measurement error.

The outcome of this test does not determine whether the preliminary estimate is a forecast or an observation with error. If the test leads to rejection, the estimate is not a well-behaved member of either category. If it is passed, it is meaningful to ask to what extent the preliminary estimate represents pure observations or a conditional forecast. A previous version of this article demonstrated that the parameter λ , or at least a lower bound for it, may be identified from the variance-covariance matrix of the system consisting of Equation (4) and the unconditional forecasting equation for y_i . This identification depends, however, on u_i being white noise. As suggested in Section 2, it may take considerable time from the release of y_i^p until y_i^* is known. Thus, for some $s > 0$, u_{i-s} may not be part of X_i , which represents the information available to the BEA at the time of the release of y_i^p . Consequently, serial correlation of u_i is not a violation of the null hypothesis for Equation (4). Since some serial correlation appears to be present empirically for all of the specifications considered in this article, the issue of identifying λ is not pursued.

Mankiw et al. (1984) presented alternative tests to determine whether the preliminary announcements are efficient forecasts or preliminary observations. In general, these tests centered on two regressions, one of y_i^* on y_i^p and X_i and one of y_i^p on y_i^* and X_i . If y_i^p is an efficient forecast, it will have a unitary coefficient in the first equation, whereas all other coefficients in that equation, including the intercept, are 0. Conversely, if y_i^p is a preliminary observation, the second regression should yield analogous results. Although these tests provide important information, they also are problematic. They test the joint hypotheses that the announcements are forecasts and that these forecasts are efficient or that the announcements are observations and that the observations are random, respectively. The test of well-behavedness based on Equation (4) is more general, because it is independent of the value of λ . If the announcements are mixtures of observations and forecasts ($0 < \lambda < 1$ or $k < 1$), both of the Mankiw–Runkle–Shapiro tests may lead to rejection, even if the forecasts are efficient and the observation errors ran-

dom. As the results in Section 6 indicate, however, spurious evidence of ill-behavedness does not seem to be the problem in this case.

4. EMPIRICAL IMPLEMENTATION AND ESTIMATION ISSUES

This article studies the preliminary estimates for quarterly growth rates of constant-dollar GNP. The growth rates are preferred to levels, not only because they receive more attention in the media, but also because considering them sidesteps the problem of non-stationarity for the level data, as pointed out by Mankiw and Shapiro (1986).

As variables reflecting information at the time of the estimate, I include a constant, the most recent estimate of GNP growth in the preceding quarter, and the median forecast from the National Bureau of Economic Research/American Statistical Association (NBER/ASA) survey of real GNP growth for the quarter for which the estimate is made.

In terms of Equation (4), the purpose of the constant is to detect potential biases. A conditional bias would be consistent with the null hypothesis if the preliminary estimate is an observation with error. That is, the conditional expectation of the revision, given the observation as well as other information, is nonzero. The unconditional expectation—given only the previously available information—should be 0, however. The way (4) is specified, a nonzero constant term reflects an unconditional bias, which is a violation of the null hypothesis.

The coefficient on the most recent estimate of lagged GNP growth is nonzero if the process of deriving the preliminary estimates relies excessively on extrapolations of recent trends. Such overreliance is likely to result, for example, in failures to identify turning points on the business cycle when such turns in fact are indicated by the available information. If such failures are prevalent, this coefficient should be negative.

The NBER/ASA median forecast of real GNP growth is intended to serve as a summary of other available information. These forecasts are not presented as a norm with which to compare the preliminary estimates, nor is it implied that the forecasts are efficient. In fact, Zarnowitz's (1985) examination of the same set of forecasts finds many signs of inefficiency. They are introduced here simply because they summarize available information and may themselves serve as a source of information to the BEA whenever the agency includes elements of forecasts in its early estimates. Whether or not such elements are included, the information should be uncorrelated with the subsequent revisions, as implied by the preceding section. The BEA may not have had direct and immediate access to the NBER/ASA survey data, but it certainly had access to the data on which the forecasts were based. Furthermore, the BEA obviously had the opportunity to conduct a similar survey of its own.

If u_t is independently and identically distributed over

time, estimation of Equation (4) by ordinary least squares (OLS) is efficient, and tests of $H_0: \beta = 0$ can be based on the OLS estimates of the variance-covariance matrix of the parameter estimates. Empirically, u_t appears to exhibit some serial correlation. As pointed out in Section 3, such serial correlation does not necessarily violate H_0 because of the lag involved in the revision process. Nonzero autocorrelations at long lags would seem suspicious, however. Thus, within the null hypothesis, a moving average (MA) process of a relatively low order seems a reasonable specification of the time series behavior of u_t .

In this context, the OLS estimate $\hat{\beta} = (X'X)^{-1}X'r$ is consistent, although not efficient. Moreover, the estimate of the variance-covariance matrix for $\hat{\beta}$ computed by the standard OLS routine is inconsistent. In the standard econometric model, in which the elements of X are exogenous, efficient estimation of β as well as consistent estimation of its variance-covariance matrix is obtained by a generalized least squares (GLS) transformation. The GLS procedure uses a polynomial in the lag operator, $\phi(L)$, to transform the data, where ϕ is constructed so that the filtered error term, $\phi(L)u_t$, is white noise.

Unfortunately, this procedure may lead to inconsistent estimation of β if the variables on the right are predetermined rather than exogenous. Although a predetermined variable is unaffected by current (and future) shocks to the economy, it has been affected by past shocks. Except for the constant, the X variables listed previously fit this description; that is, $X_t \perp u_s$ for $s \geq t$, but not necessarily for $s < t$. Now, $\phi(L)X_t$ is a function of X_t , but $\phi(L)u_t$ is a function of u_s for $s < t$. The resulting potential correlation between $\phi(L)X_t$ and $\phi(L)u_t$ explains the inconsistency of GLS. This problem, which may arise in many economic models, was pointed out by Flood and Garber (1980).

Recent econometric literature has suggested two solutions. Hayashi and Sims (1983) proposed the "forward filter" $\phi(L^{-1})$ as an alternative to the GLS transformation. This filter whitens u_t as well. As in the GLS transformation, $\phi(L^{-1})X_t$ is correlated with $\phi(L^{-1})u_t$. Since $\phi(L^{-1})u_t$ is a function only of u_s ($s \geq t$), however, the untransformed X_t is uncorrelated with the transformed error term and thus is eligible as an instrument.

The alternative solution is based on the GMM. The various aspects of this method have been analyzed by Hansen (1982), Cragg (1982), Gallant (1985), and others. In the present context, the GMM point estimates are identical to the ones obtained by OLS; however, GMM provides a consistent estimate of their true variance-covariance matrix.

The choice between GMM and the forward filter is not always clear. Either method is consistent, neither method is efficient in an absolute sense, and their relative efficiency depends on the problem at hand. GMM is chosen here because the consistency of the GMM estimate of the variance-covariance matrix, and hence

the validity of inference, is robust with respect to conditional heteroscedasticity (i.e., $\text{var } u_t$ may be a function of X_t). The forward filter method does not have this property without further modification.

The correction of the variance-covariance matrix for $\hat{\beta}$ takes the following form. Under general conditions, it can be shown that the vector $T^{1/2}(\hat{\beta} - \beta)$, where T is the sample size, is asymptotically normally distributed with mean 0 and variance-covariance matrix

$$\text{plim } T^2(X'X)^{-1}M(X'X)^{-1},$$

where the positive definite matrix M is defined as

$$M = \lim T^{-1}E(X'uu'X)$$

with elements

$$m_{ij} = \lim T^{-1} \sum_{t=1}^T \sum_{k=1-t}^{T-t} E(x_{it}x_{jt+k}u_tu_{t+k}),$$

where x_{it} , $x_{j,t+k}$ denote elements of the data matrix X for the variables on the right.

The form of m_{ij} reveals that, if the time series $\{u_tX_t\}$ is stationary, M is the spectral density of this series evaluated at 0. Consequently, similar problems arise in estimating M as in estimating spectral densities. For further discussion, the reader is referred to the detailed analysis of spectral density estimation in Anderson (1971, chap. 9) and the application to GMM estimation in Gallant (1985).

If the population moments in the formula for m_{ij} are replaced by sample moments, the resulting estimator is not, in general, consistent, because the rapid increase in the number of summands (T^2 rather than T) as the sample grows prevents the variance of the estimator from approaching 0. This problem can be bypassed, however, by the assumption that $\{u_t\}$ is a moving average process, where the order of the process, denoted by R , does not depend on T . This assumption suggests the estimators

$$\hat{m}_{ij} = T^{-1} \sum_{t=1}^T \sum_{k=a}^b x_{it}x_{j,t+k}\hat{u}_t\hat{u}_{t+k},$$

where $a = \max(-R, 1 - t)$ and $b = \min(R, T - t)$.

These estimators are consistent, essentially because the number of summands grows with T rather than T^2 . Unfortunately, however, the implied estimate of M is not guaranteed to be positive definite. Positive definiteness can be obtained by using

$$\hat{m}_{ij} = T^{-1} \sum_{t=1}^T \sum_{k=a}^b w_k x_{it}x_{j,t+k}\hat{u}_t\hat{u}_{t+k}, \quad (5)$$

where w_k are the Parzen (1961) weights

$$\begin{aligned} w_k &= 1 - 6|k/(R+1)|^2 + 6|k/(R+1)|^3, \\ &0 \leq |k| \leq (R+1)/2 \\ &= 2[1 - |k/(R+1)|]^3, \\ &(R+1)/2 \leq |k| \leq R+1. \end{aligned}$$

On the other hand, application of the Parzen weights interferes with consistency under the assumption that R is fixed independently of T , mainly because the weight for a given k does not approach unity as T grows.

Both consistency and positive definiteness are obtained by using a formula of the same form as (5), with unitary weights if the estimated matrix is positive definite and with w_k equal to the Parzen weights otherwise. It is obvious that this procedure ensures positive definiteness. Consistency is ensured because (5), with $w_k = 1$, is consistent. Thus, asymptotically, the matrix estimated with unitary weights is arbitrarily close to the true matrix M and hence (by continuity) will share its definiteness property, except on a set of probability 0. This means that, asymptotically, the Parzen weights will not be applied except with probability 0, which implies consistency.

Abel and Mishkin (1983) analyzed the validity of orthogonality tests of the type $H_0: \beta = 0$ for Equation (4) when variables representing some of the relevant information are excluded from the equation. They found that rejections of the null hypothesis remain valid under this type of misspecification. The reasoning behind this result essentially goes as follows. Suppose β is the coefficient vector for the completely specified equation. Then, according to standard misspecification analysis, the probability limits of the coefficients actually estimated are linear combinations of the true values of the elements of β . Since $\beta = 0$ under H_0 , a rejection of the hypothesis that the estimated coefficients are 0 also implies a rejection of the correct H_0 .

This result has two implications for this study. First, if the hypothesis of zero coefficients for Equation (4) is rejected for the elements of X_t listed previously, this finding is a valid rejection of the theoretical null hypothesis even if X_t leaves out relevant information. Second, if the hypothesis is rejected if only a subset of the variables listed previously are included in the equation, this rejection remains valid as well.

5. THE DATA

In support of Parker's (1984) revision study, the BEA staff prepared time series for all of the vintages of the GNP estimates. The series go back to the second quarter of 1968, except for the 75-day estimate, for which regular releases started with the second quarter of 1974. Prior to September 1983, the flash estimates were computed but not published; however, the previously unpublished estimates are included in the data used here. The data have been normalized by the BEA for changes in classifications, definitions, and choice of evaluation year, and Parker graciously made them available.

There is some question as to the definition of the final real GNP data. Since the constant-dollar estimates are revised indefinitely as the evaluation years are changed, no estimate is ever completely final. This article follows Parker by simply treating as final the latest

figure available at the time the research was done. For all quarters through 1980, this means the 1980 benchmark estimate. For 1981, 1982, and 1983, it is the third, second, and first July revision, respectively. For the first quarter of 1984, it is the 105-day estimate. For the remaining observations, the 75-day estimate is the latest one available. It is used as the final estimate in the study of the revisions of the flash, 15-day, and 45-day estimates. For revisions in the 75-day estimate, the first quarter of 1984 defines the end of the sample. For the three earlier vintages, the sample is defined by the availability of the NBER/ASA forecast data, which will be discussed further.

The results should not be sensitive to the ambiguity of the definition of the final estimate. If the estimates that I treat as final do in fact measure the true value with a random observation error, or if they forecast the true value with a random prediction error, the form of Equation (4) remains intact, including the null hypothesis. The only difference will be an added component to the error term that should be uncorrelated with the variables on the right. On the other hand, a rejection of the null hypothesis could be caused by a systematic error in the final estimate. Although such a rejection correctly would point to problems in the process generating GNP estimates, it might be incorrect to ascribe the problem to the preliminary rather than to the final estimate.

Because of the normalizations for definitions, classifications, and changes in evaluation year, the data used in the study are not exactly the same as the estimates actually released at the time. These normalizations particularly affect the early part of the sample. Their effect on the results should be minor, however. The normalizations consist of the addition of an equal amount to each of the level estimates. Therefore, the impact on growth rates should be small.

The more recent estimates have been subjected to fewer and less thorough revisions. Therefore, the revisions of the more recent data should be expected to be smaller even if the quality of the estimates remain constant over time. Since the data from both the early and the late part of the sample suffer from some problems, it is not clear which part gives rise to the more reliable results. As the next section shows, however, the results from different subsamples are not radically different in practice. A better test for the late part of the sample may be possible when suitable data from the 1985 benchmark revision become available. Use of the benchmark data are complicated by the change in evaluation year. A special data series currently under preparation at the BEA that incorporates the information in the benchmark revision but uses 1972 as the evaluation year should help overcome this complication. I expect to examine these revisions in a future article.

Some commentators on an earlier draft suggested that I compare the initial estimates with a less radical

revision as the "final" estimate, such as the third July revision. Although such an undertaking could be the subject of a separate article, I would argue that this test would be too lenient. Only the benchmark revisions seem to use the detailed information on which the national income accounts reasonably can be expected to be based. The only reason the preliminary estimates should differ systematically from this target should be changes in definitions and other methodical changes. Since the data are normalized for these changes, at least in principle, the procedure used here seems appropriate.

The NBER made the data for the NBER/ASA survey of economic forecasts available. This survey presents individual forecasts for the fourth quarter of 1968 through the fourth quarter of 1984. Starting in 1981, the respondents to the survey were asked to state their forecasts for real GNP. Prior to this time, the survey asked for forecasts for nominal GNP and the GNP deflator. I used the explicit forecast for real GNP when available; otherwise, I used the forecast implicit in nominal GNP and the deflator. The survey was conducted during the second month of each quarter so that all of the information available to the forecasters was available for the construction of the flash and later estimates as well. The forecasts are given in level form; however, since the responses also include the levels for the last quarter before the survey, the conversion to growth rates is straightforward.

My data series consists of the median of the forecasts for the quarter in which the survey was taken. Because of some suspicious outliers, I decided to use the median rather than the mean forecast. Experiments with the mean forecast resulted in quite similar results, however. The number and identity of the responses varied from quarter to quarter, quite possibly in a nonrandom manner.

The data used for this article are available from me on request.

6. RESULTS

The empirical results for Equation (4) are presented in Tables 1–4 for each of the GNP vintages, respectively. The estimations were carried out for the full sample 1968.4–1984.4, as well as for each of the two halves (1968.4–1976.4 and 1977.1–1984.4) of this sample. The equations for the full and the first half sample for these vintages use the lagged 45-day estimate as the most recent estimate of the previous-quarter growth. For the second half, the 75-day estimate was available and is used instead.

The sample for the 75-day estimate is truncated on both ends because of the limited availability of the data. I start this sample in 1974.3 to include the lagged value of this estimate as the most recent estimate of previous quarter growth. This sample of 39 observations is too short to be divided into subsamples.

Table 1. Flash Estimate: Parameter Estimates and Test Results for Equation (4)

Sample	Constant	45-day estimate, lagged	75-day estimate, lagged	NBER/ASA forecast	Approximate F statistic, $H_0: \beta = 0$
1968.4–1984.4	.54 (.21)				6.61 (.012)
	.87 (.27)	-.13 (.04)			6.16 (.004)
	.59 (.24)	-.24 (.08)		.24 (.09)	5.63 (.002)
1968.4–1976.4	.24 (.35)				.47 (.496)
	.72 (.37)	-.19 (.05)			7.78 (.002)
	.26 (.36)	-.35 (.09)		.31 (.15)	23.88 (.000)
1977.1–1984.4	.85 (.14)				36.24 (.000)
	1.04 (.31)		-.07 (.06)		7.46 (.002)
P	.86 (.38)		-.18 (.09)	.25 (.10)	4.16 (.014)

NOTE: The estimation method is generalized method of moments. Standard errors and marginal significance levels are in parentheses. Parzen weights are applied in specifications marked P.

Preliminary analyses of the OLS residuals were used to determine the degree R of the MA process of the error term. In most cases, first- and second-order autocorrelation was weak, but third-order autocorrelation was found to be substantial in some cases, especially for the second half of the sample for the flash, 15-day, and 45-day estimates. Higher-order autocorrelation was weak. Setting $R = 3$ would lead to an estimate of M that would account well for the observed correlations

with unitary weights in Formula (5). The Parzen weight for $k = 3$ when $R = 3$ is, however, only .03, which seems unfortunate when most of the autocorrelation is found at the third lag. Raising R to 4 ameliorated this problem and led to fewer cases for which the Parzen weights needed to be imposed. Experiments with $R = 5$ led to very similar results. Consequently, I used $R = 4$ for the final estimations for all specifications and all samples.

Table 2. 15-Day Estimate: Parameter Estimates and Test Results for Equation (4)

Sample	Constant	45-day estimate, lagged	75-day estimate, lagged	NBER/ASA forecast	Approximate F statistic, $H_0: \beta = 0$
1968.4–1984.4	.43 (.17)				6.17 (.016)
	.75 (.22)	-.13 (.06)			5.75 (.005)
	.55 (.19)	-.21 (.10)		.17 (.11)	5.06 (.003)
1968.4–1976.4	.25 (.32)				.60 (.445)
	.81 (.22)	-.22 (.05)			12.37 (.000)
	.40 (.35)	-.37 (.08)		.28 (.15)	128.37 (.000)
1977.1–1984.4	.61 (.16)				14.33 (.001)
	.69 (.21)		-.03 (.06)		7.31 (.003)
P	.58 (.37)		-.09 (.10)	.15 (.12)	2.08 (.124)

NOTE: The estimation method is generalized method of moments. Standard errors and marginal significance levels are in parentheses. Parzen weights are applied in specifications marked P.

Table 3. 45-Day Estimate: Parameter Estimates and Test Results for Equation (4)

Sample	Constant	45-day estimate, lagged	75-day estimate, lagged	NBER/ASA forecast	Approximate F statistic, $H_0: \beta = 0$
1968.4–1984.4	.30 (.15)				3.80 (.055)
	.59 (.18)	-.11 (.05)			5.17 (.008)
	.46 (.15)	-.17 (.08)		.11 (.10)	4.62 (.006)
1968.4–1976.4	.18 (.28)				.42 (.520)
	.69 (.19)	-.20 (.05)			13.77 (.000)
	.51 (.25)	-.26 (.10)		.12 (.15)	26.01 (.000)
1977.1–1984.4	.43 (.17)				6.34 (.017)
	.47 (.19)		-.02 (.05)		3.45 (.045)
	.37 (.14)		-.08 (.10)	.14 (.11)	2.97 (.048)

NOTE: The estimation method is generalized method of moments. Standard errors and marginal significance levels are in parentheses.

The GMM estimate of the variance-covariance matrix can be used to construct Wald statistics for the null hypothesis in the usual fashion. Only the asymptotic properties of these statistics are known. In particular, it is not certain that the F distribution (after correction for degrees of freedom) provides a better approximation to the true small-sample distribution than the χ^2 distribution, which holds asymptotically. Nevertheless, since using the χ^2 distribution always leads to more frequent rejections, I use the F distribution simply as a more conservative testing strategy.

Table 1 presents the results for the flash estimate. For the full sample, the null hypothesis is rejected on the 5% level for all of the specifications. The results indicate an unconditional downward bias of about one-half of a percentage point when measured on the basis of the full sample. The most recent estimate of lagged growth enters significantly with a negative sign, suggesting a systematic failure to identify turning points. This picture gets even worse when other information is

taken into account, as reflected in the NBER/ASA forecasts; the flash estimate fails to identify those turning points that are predicted correctly by the outside forecasters. Furthermore, there is a significant correlation between the revisions and the forecasts. This relationship is not significant as a simple (noncentral) correlation; however, when included in a regression together with a constant and the lagged growth estimate, the forecast variable has a coefficient that is significantly different from 0.

The flash estimate was not officially released to the public until September 1983, although it was circulated for internal use among federal agencies and frequently leaked to the public. In this situation, it is reasonable to ask if its performance has changed over time. The pressure of the public's scrutiny might, on the one hand, lead to efforts to improve the estimates. On the other hand, it also might lead to a reluctance to release estimates that deviate too sharply from recent trends.

The observations starting in September 1983 are too few to form a meaningful sample. As an alternative, Table 1 includes the results for the two halves of the sample. Some changes over time are worth noticing. First, the bias clearly seems to have worsened over the sampling period, rising from about a quarter of a percentage point in the early period to close to a full percentage point later. On the other hand, the overreliance on extrapolations of recent growth rates seems to have diminished substantially. The correlation with independent forecasts also is reduced a little. This correlation, together with the substantial bias, is ample reason for rejection of the null hypothesis for the second half of the sample, however.

Public comments on the flash estimate sometimes

Table 4. 75-Day Estimate: Parameter Estimates and Test Results for Equation (4)

Constant	75-day estimate, lagged	NBER/ASA forecast	Approximate F statistic, $H_0: \beta = 0$
.18 (.26)			.46 (.500)
.32 (.32)	-.06 (.06)		.61 (.550)
.30 (.31)	-.08 (.08)	.03 (.11)	.40 (.76)

NOTE: The estimation method is generalized method of moments. Standard errors and marginal significance levels are in parentheses. The sample is from 1974.3–1984.1.

imply a much poorer performance for this estimate than the later vintages. For example, a study by King (1982) found that about half of 33 prominent business economists surveyed rate the 15-day estimate as "most important," but only one-fifth give the same rating to the flash. The recent decision to discontinue publication of this estimate appears to have been influenced by such comments. My results, however, do not indicate any radical difference in quality between the flash and the 15-day estimates, as indicated by the results shown in Table 2. For the full sample, the bias is a little lower, but it is just as significant. The coefficients for recent growth and the NBER/ASA forecasts are almost as large with similar standard errors.

For the first half of the sample, the performance is very similar to the flash estimate. The second half is also very similar, but the 15-day estimate cuts a little off the bias and the coefficient for the NBER/ASA forecasts is insignificant. In fact, the F statistic for the full specification for the second half is insignificant at the 5% level. Since rejections of the null hypothesis are valid even if some variables on the right are excluded, however, the bias is strong enough by itself to reject well-behavedness forcefully.

The second half of the sample consists of only 32 observations. This sample size is on the low side for application of asymptotic distribution theory, which forms the basis for the GMM method. Thus it is possible that the marginal significance levels are exaggerated for each of the subsamples. The margin for rejection of the hypothesis of no bias for the second half is very wide, however. It does not seem likely that small-sample corrections should change the results substantially.

There is an additional reason to be skeptical of favorable results for the most recent data. Most revision studies have shown improvements over time. Unfortunately, however, these improvements do not necessarily reflect real improvements in the collection and processing of data. Instead, they may reflect the fact that the more recent data have been subject to fewer and less comprehensive revisions. Thus it is necessary to warn against premature conclusions about improvements.

Previous terminology referred to the 15-day estimate as "preliminary" and the 45-day estimate as "revised." This terminology appears to reveal an expectation that the 45-day estimate somehow performs better. Judging from the results in Table 3, it performs a little better relative to the criteria used here, but not a lot better on the whole. For the full sample, the bias is smaller than for the flash and the 15-day estimates. It is borderline significant at the 5% level. Lagged growth is significant for the full sample, however, and this variable, together with the constant, leads to a highly convincing rejection of the null hypothesis.

For the first half of the sample, the bias is small and insignificant for this vintage also. The results for the other variables, especially the lagged growth rate, war-

rant unambiguous rejection of the null hypothesis, however. The results for the second half follow the pattern of the 15-day estimate. Again the bias is significant, although smaller than for the earlier announcements. The other variables are not significant, although the null hypothesis is rejected for all specifications.

The results for the 75-day estimate permit more optimistic conclusions. As shown in Table 4, the data do not even come close to rejecting the null hypothesis on any reasonable significance level. Although the sign of the constant term persists, it is much smaller and not statistically significant. The other coefficients are insignificant economically as well as statistically. Overall, the null hypothesis is very far from rejection for all specifications.

The substantial difference between the results for the 45-day and 75-day estimates is remarkable because the revisions between them tend to be smaller than the revisions between the other estimates. Thus the quality of these revisions must be more important than their quantity. It seems that the longer period available for preparation of the 75-day estimate provides for a better-quality estimate. On the other hand, this finding is tempered somewhat by the fact that the short sample of available 75-day estimates has been subjected to few thorough revisions. Finally, it remains possible that other variables than those used here can predict the revisions of the 75-day estimate. Nevertheless, the inclusion of the Index of Leading Economic Indicators in a previous version of this article produced no different results.

7. COMMENTS AND CONCLUSIONS

The results of this article suggest that the stochastic behavior of the first three preliminary GNP estimates is highly unsatisfactory. They behave neither as efficient forecasts nor as observations with random errors. They show a systematic downward bias. Furthermore, their tendency to depend too much on extrapolations of recent trends produces a tendency to miss cyclical turning points when such turns in fact are indicated by the available information. These problems are not confined to the flash estimate, which commonly has been considered unreliable and recently was discontinued. They are shared by the 15-day and 45-day estimates, although some improvement can be noticed for each successive estimate. In contrast, the 75-day estimate appears to be well behaved. Since the 75-day estimate was not issued regularly until 1974, however, the evidence for this estimate is more limited.

These results are subject to some caveats. The data for the early part of the sample have been normalized to account for changes in classifications, definitions, and evaluation year. Although the effects of these normalizations are much smaller for the growth rates than for the level data, it is conceivable that they have contributed to the impression of poor behavior. The more recent data have seen less normalization, simply be-

cause at the time the research was done no major revisions had taken place since the 1980 benchmark revision. By the same token, however, one must expect the more recent data to have the appearance of superior behavior. Considering that the analysis finds significant flaws in this portion of the sample as well, the overall conclusions of the article seem very strong. The 1985 benchmark revision will provide useful material for further research.

Although the main results of this article may be disappointing, they are not necessarily difficult to explain. Every GNP announcement by the BEA is followed with intense interest by the private as well as by the public sector. The criticism for "false signals" typically is harsh. In such an environment, the BEA must be expected to follow a conservative strategy.

This outside pressure seems a sufficient explanation for the tendency to rely excessively on extrapolations of recent trends. Although the criticism for missing major changes may be hard, the criticism for issuing false signals may be even harder. Extrapolating recent trends would seem a reasonably safe bet for the BEA.

Similarly, the downward bias may be interpreted as a reluctance to issue overly optimistic estimates only to be criticized for false optimism later. Furthermore, it may not be necessary for the BEA to release large numbers in order to signal strong growth. Suppose, for example, that the available observations and other information indicate 9% growth. By releasing an estimate of, say, 6%, the BEA can both signal strong growth and hedge against a potential embarrassment should the current signals turn out to be misleading.

One might expect this argument to be symmetric; that is, the BEA should be as reluctant to issue very small (or negative) numbers as very large numbers. Perhaps the observed asymmetry is related to the asymmetry in the economy's performance. Economic decline tends to be concentrated in a few dramatic quarters in which the weakness of the economy is widely apparent. Thus, issuing low numbers for these quarters may not be quite as risky. Nevertheless, this explanation does not seem convincing in all cases. A certain puzzle in regard to the bias seems to remain.

A contribution to the improvement of the methods for estimating early GNP figures would be a desirable outcome of this article. The results imply some adjustments that could make the revisions less predictable, but that was not the article's primary purpose. Rather, it was to contribute to the user's understanding of the early GNP estimates. The findings seem to go quite some distance toward this goal.

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